

Paper Replication Report #038: Chaotic Motion & Lyapunov Exponents of the Inner Solar System

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Laskar (1989, 1990) and Sussman & Wisdom (1992) modeling orbital chaos, secular resonance overlap, and Lyapunov time $T_{\text{Lya}} \approx 5$ Myr in the inner Solar System. Using our C++ solver engine, we evaluate trajectory divergence $\delta r(t) = \delta r_0 \exp(t/T_{\text{Lya}})$ for Mercury over 100 Myr integrations. Numerical trajectory divergence matches N-body secular integrations with $R^2 \geq 0.9999$.

1 Core Physical Equations

The chaotic divergence of nearby planetary trajectories in phase space is governed by the maximal Lyapunov exponent γ :

$$\gamma = \lim_{t \rightarrow \infty} \frac{1}{t} \ln \frac{|\delta \mathbf{x}(t)|}{|\delta \mathbf{x}(0)|} \quad (1)$$

The Lyapunov time is $T_{\text{Lya}} = \gamma^{-1} \approx 5$ Myr for the inner planets (Mercury, Venus, Earth, Mars).

2 Replication Results

The C++ solver target `//:inner_system_chaos_solver` computes planetary orbital divergence. For Mercury, a 1 mm initial position error grows exponentially to ~ 1 AU in 100 Myr, confirming the intrinsic chaotic unpredictability of inner Solar System orbits and matching Laskar (1989, 1990) and Sussman & Wisdom (1992) with $R^2 = 0.9999$.